

🌐 Live Web Application: <https://roamcraft.onrender.com>
📁 GitHub Repository: <https://github.com/sagun-glitch/RoamCraft>

1. Executive Summary

RoamCraft is an AI-driven vacation planning platform built with **Node.js**, **Express**, and **Google Gemini**. Featuring a **Split-Screen Expedition Studio UI**, it streams personalized day-by-day itineraries live via **Server-Sent Events (SSE)** based on user destination, duration, budget tier, and custom activity preferences.

2. Key Highlights & Features

- **Split-Screen Expedition Studio Paradigm:** Engineered with a dual-canvas layout—left control panel for interactive trip parameter building and right stage canvas for real-time itinerary stream rendering.
- **Real-Time AI Streaming:** Delivers instant markdown-formatted itinerary updates chunk-by-chunk using Server-Sent Events (SSE) without long loading delays.
- **Precision Personalization:** Tailors daily itineraries based on budget tier (Budget-friendly, Moderate, Luxury) and custom interest preferences.
- **Interactive Day-by-Day Dashboard:** Renders structured day blocks with morning/afternoon/evening schedules, interactive packing checkboxes, and local travel tips.
- **Cyber Charcoal & Emerald Slate Theme:** High-contrast, clean UI built with Outfit & Fira Code fonts—strictly NO gradients and NO glassmorphism.

3. Technical Stack & Architecture

Layer	Technology	Details
Live Hosting	Render Cloud Platform	Deployed live at https://roamcraft.onrender.com
Backend Engine	Node.js / Express.js	Manages REST endpoints, static file serving, and SSE streaming pipeline

Layer	Technology	Details
Generative AI	Google Gemini API	High-speed generative AI model configured for structured streaming output
Frontend UI	HTML5 / Vanilla CSS / ES6+ JS	Expedition Studio split-screen interface built with Outfit, Inter & Fira Code fonts, and Lucide Icons
Streaming Protocol	Server-Sent Events (SSE)	HTTP streaming protocol for progressive chunk rendering in client browser

4. API Endpoint Reference

POST `/api/generate-itinerary`

Request Payload (JSON):

```
{
  "destination": "Kyoto, Japan",
  "days": 3,
  "budget": "Moderate",
  "interests": "temples, tea houses, ramen"
}
```

5. Local Development & Deployment

- Clone repository: `git clone https://github.com/sagun-glitch/RoamCraft.git`
- Install dependencies: `npm install`
- Configure environment file `.env` with `GEMINI_API_KEY=YOUR_KEY`
- Start server: `npm start` (Open `http://localhost:3000` or visit `https://roamcraft.onrender.com`)

6. Metadata & Credits

- **Project Name:** RoamCraft
- **Author / Owner:** Sagun Mardi (`sagun-glitch`)
- **Live Application:** `https://roamcraft.onrender.com`
- **Repository:** `https://github.com/sagun-glitch/RoamCraft`